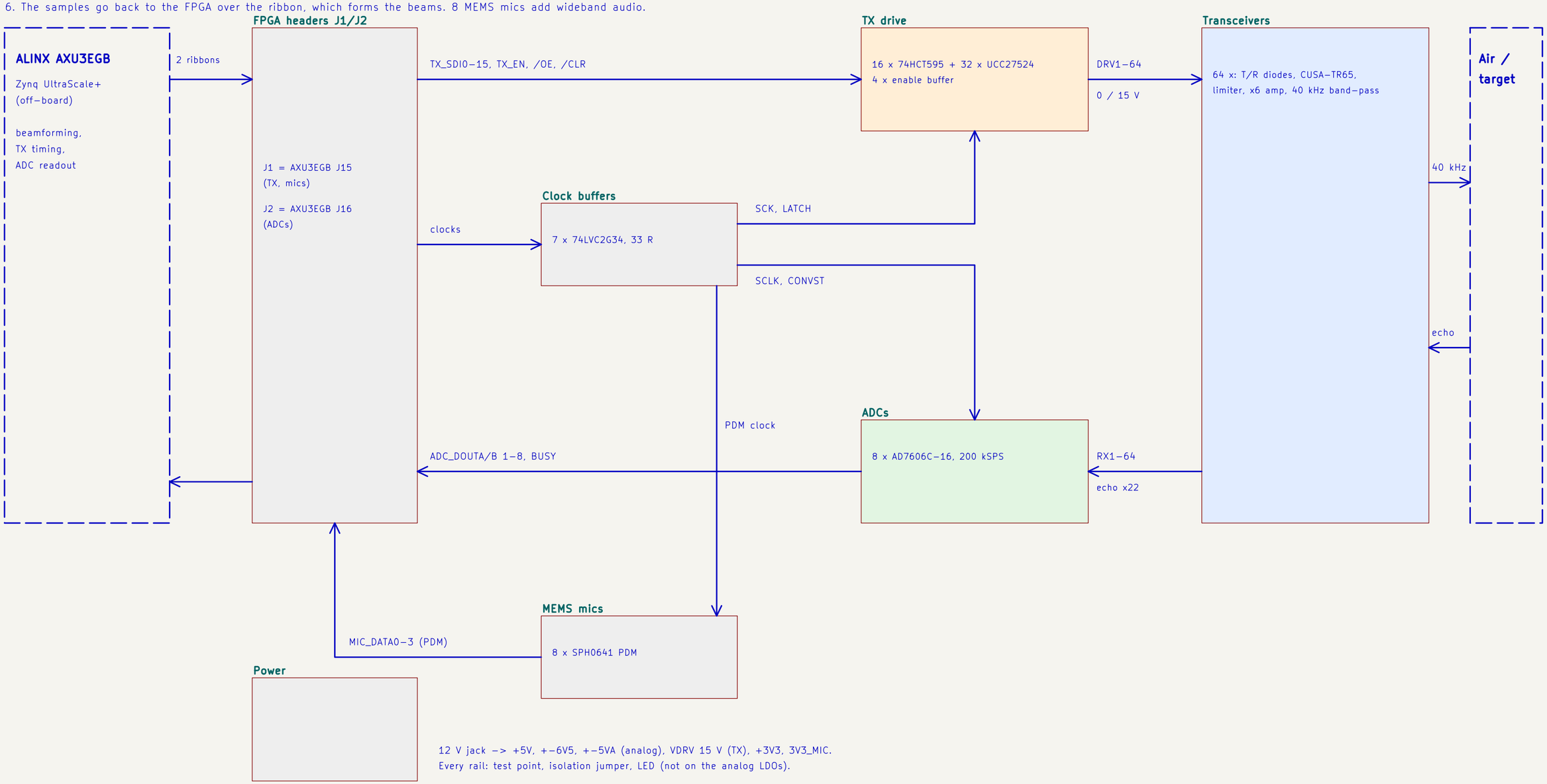
Hybrid 8x8 ultrasonic AESA: how the signal flows

1. The FPGA streams one TX bit per element (16 serial lines, 10 MHz) and the clocks.
2. TX drive: shift registers latch the bits; gate drivers switch each element between 0 and 15 V.
3. Transceivers: the 64 CUSA-TR65 elements emit 40 kHz; echoes return to the same elements.
4. The diode T/R switch and limiter protect the receiver; amplifier + 40 kHz band-pass give x22.
5. ADCs: 8 x AD7606C-16 sample all 64 channels at 200 kSPS (raw waveform, amplitude and phase).
6. The samples go back to the FPGA over the ribbon, which forms the beams. 8 MEMS mics add wideband audio.



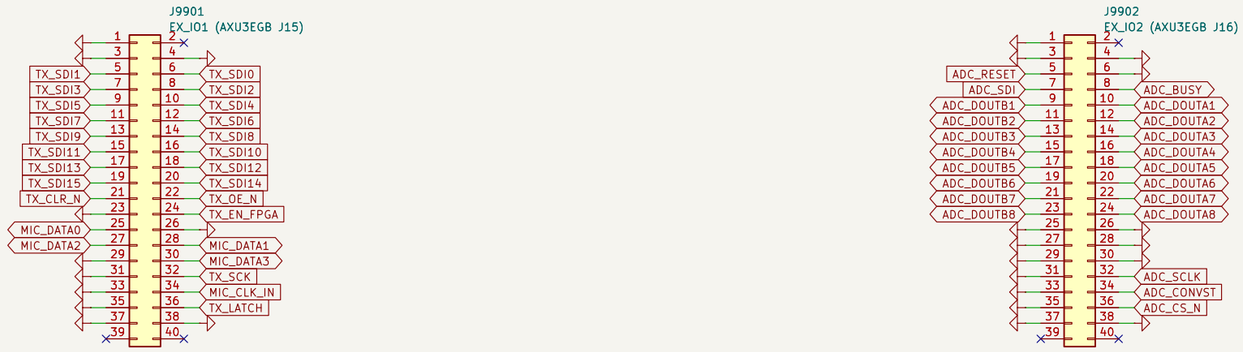
Click a block to open it. Repeated parts are one sheet each: channel x64, adc x8, tx_group x16, en_buf x4, clk_buf x7, mic x8.

Hybrid 8x8 ultrasonic AESA		
Sheet: /		
File: aesa.kicad_sch		
Title: Hybrid 8x8 ultrasonic AESA: block diagram		
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/115

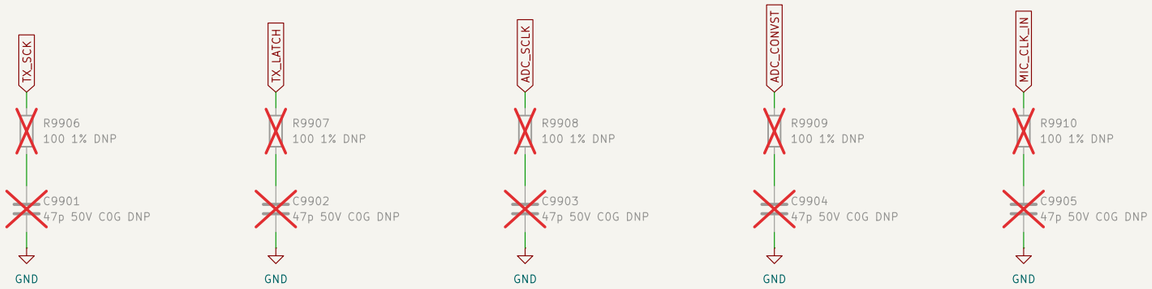
AXU3EGB expansion headers (40-pin IDC, 2.54 mm). J1 = board J15 (TX, mics), J2 = board J16 (ADCs).

Pin 2 (+5V) and 39/40 (+3.3V) of the FPGA board are left open. Clocks sit between GND conductors (pins 31–37).

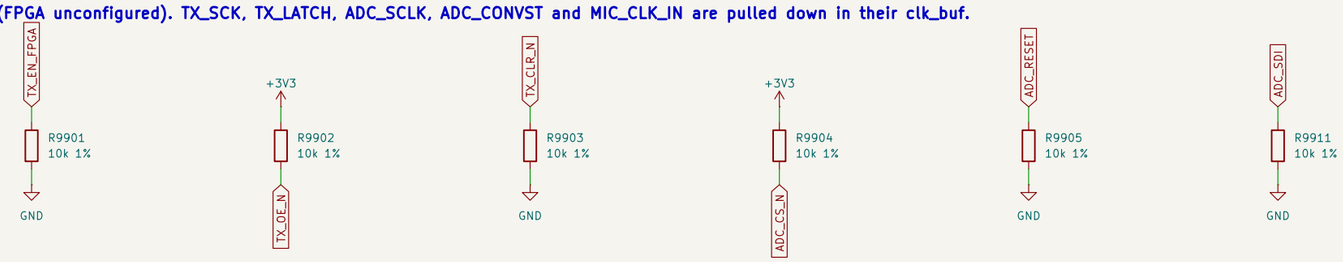
Spare FPGA pins tied to GND must stay unused (inputs) in the XDC.



FPGA ribbon outputs: DRIVE 4, SLEW SLOW (near source-matched to the 100-ohm ribbon). DNP fallback: RC end termination 100 R + 47 pF per clock, fit only with DRIVE 8.



Idle pulls (FPGA unconfigured). TX_SCK, TX_LATCH, ADC_SCLK, ADC_CONVST and MIC_CLK_IN are pulled down in their clk_buf.



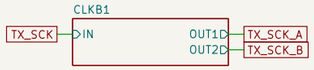
Hybrid 8x8 ultrasonic AESA

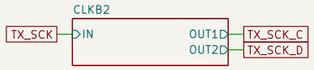
Sheet: /FPGA headers J1/J2/
File: headers.kicad_sch

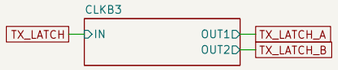
Title: AXU3EGB headers J15/J16, idle pulls

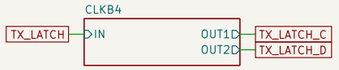
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 2/115

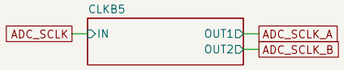
Clock buffers. Each FPGA clock enters one or two clk_buf cells; every output drives at most 4 loads through 33 R.

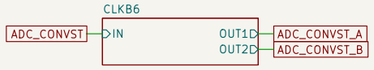














Hybrid 8x8 ultrasonic AESA

Sheet: /Clock buffers/
File: clocks.kicad_sch

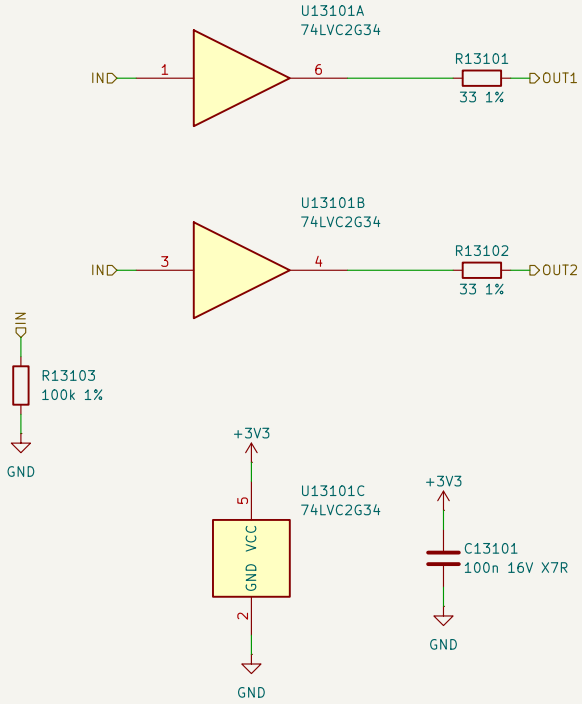
Title: Clock buffers: 7 x 74LVC2G34 at the FPGA headers

Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 3/115

Clock fan-out: 74LVC2G34 at the header, both gates on IN, 33 R series termination per output

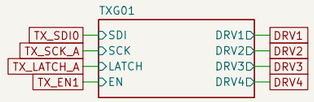
Each output drives at most 4 loads (series termination works for a lumped load at the far end).

100 k pull-down keeps the clock low while the FPGA is unconfigured.



Hybrid 8x8 ultrasonic AESA		
Sheet: /Clock buffers/CLKB1/		
File: clk_buf.kicad_sch		
Title: Clock buffer (x7): 74LVC2G34, 33 R outputs		
Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 4/115

TX drive. The FPGA streams 16 serial lines (TX_SDI0–15) into 16 tx_group cells; each drives 4 elements (DRV). TX_EN_FPGA reaches the drivers through 4 en_buf cells.























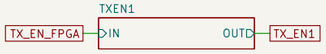


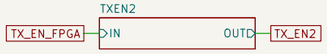


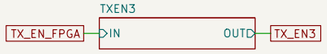














Hybrid 8x8 ultrasonic AESA

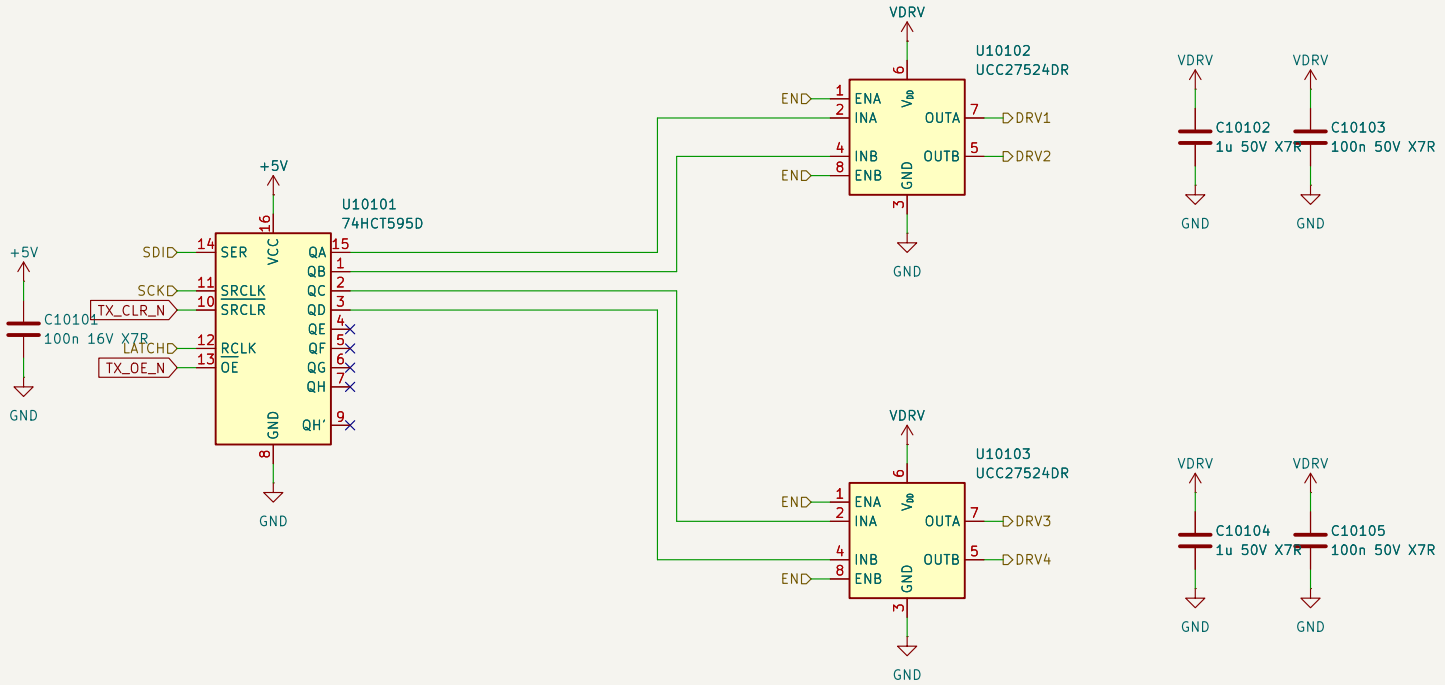
Sheet: /TX drive/
File: tx.kicad_sch

Title: TX drive: 16 shift–register/driver groups, 4 enable buffers

Size: A3	Date: 2026–10–20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 11/115

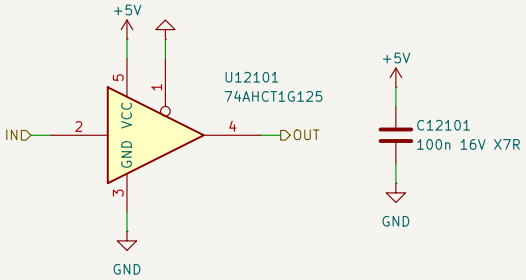
TX group: 4 elements. 74HCT595 QA–QD -> UCC27524 inputs; QE–QH unused (16 data lines, 10 MHz SRCLK)

EN from one 74AHCT1G125 per 8 drivers (en_buf). SCK/LATCH from a clk_buf output (<= 4 loads).



Hybrid 8x8 ultrasonic AESA		
Sheet: /TX drive/TXG01/		
File: tx_group.kicad_sch		
Title: TX group (x16): 74HCT595 + 2 x UCC27524, 4 elements		
Size: A4	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 12/115

TX enable buffer: 74AHCT1G125, 3.3 V in (TTL threshold), 5 V out to 8 UCC27524 (16 EN pins)



Hybrid 8x8 ultrasonic AESA

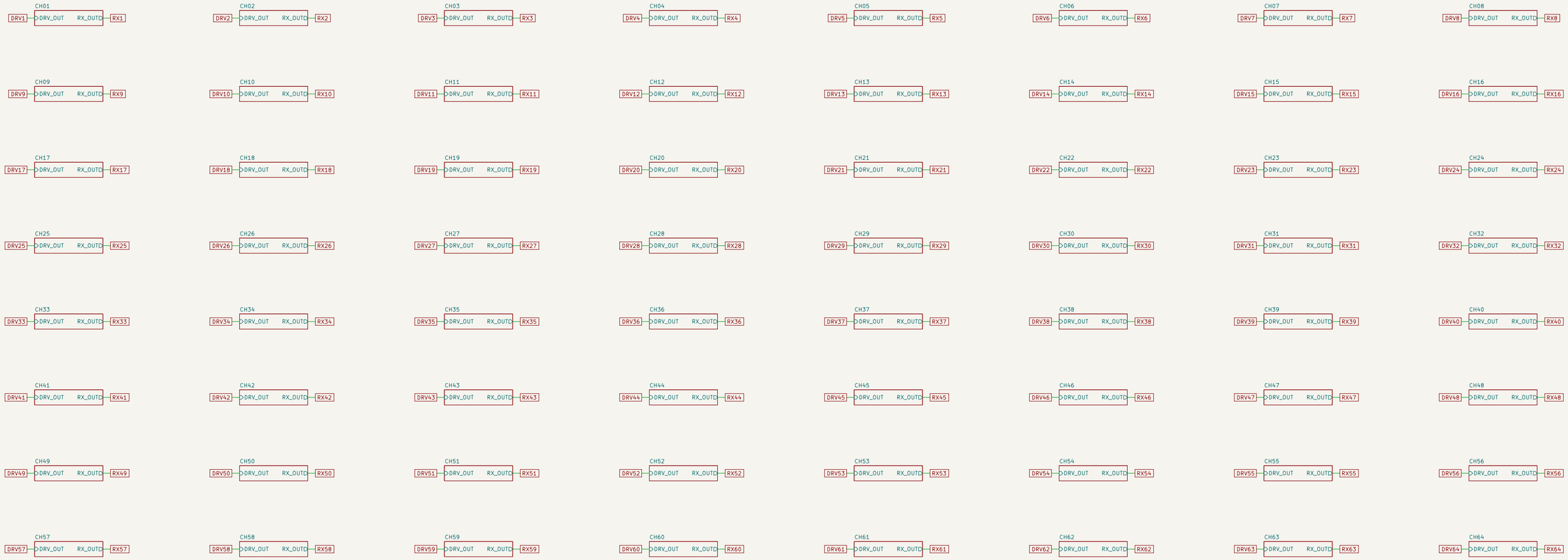
Sheet: /TX drive/TXEN1/

File: en_buf.kicad_sch

Title: TX enable buffer (x4): 74AHCT1G125

Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 28/115

Transceiver array. Channel n = element n of the 8 x 8 grid, row-major. Each channel takes DRVn from the TX drive and returns its amplified echo RXn to the ADCs.



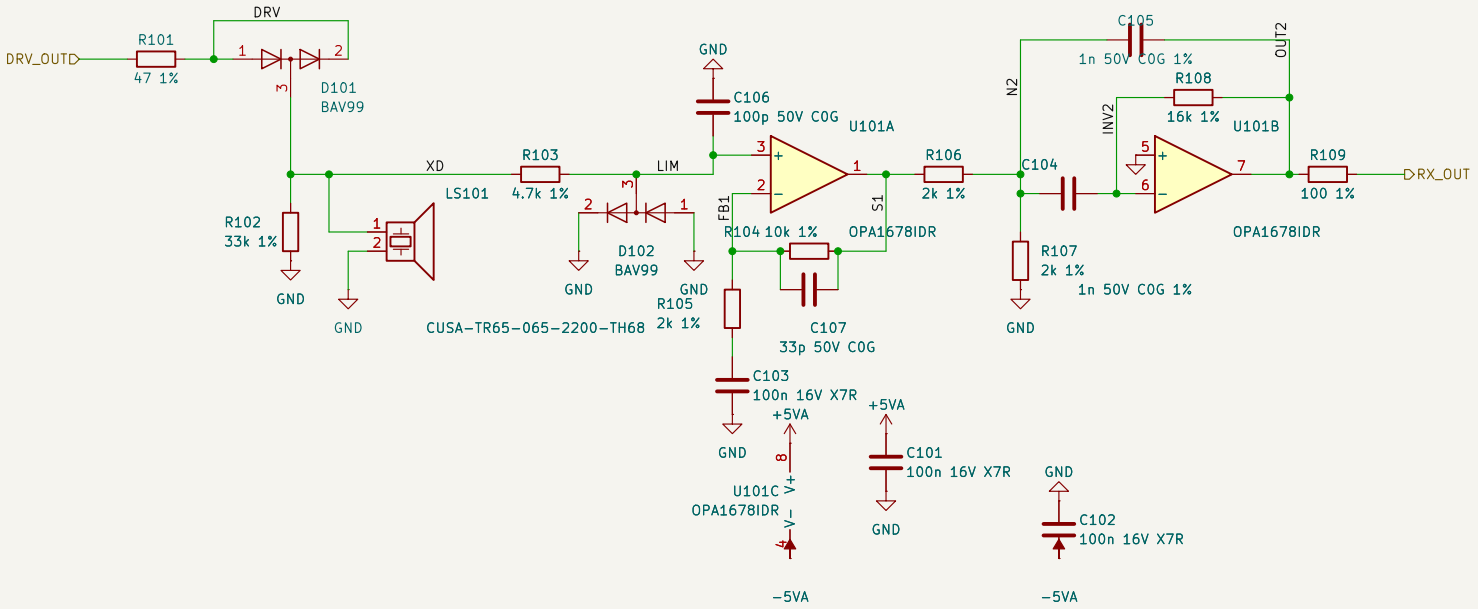
Hybrid 8x8 ultrasonic AESA

Sheet: /Transceivers/
File: array.kicad_sch

Title: Transceivers: 64 channels (8 x 8 grid)

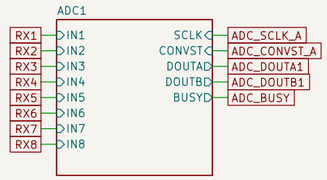
Size: A2	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 32/115

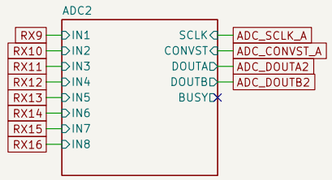
Transceiver channel (one of 64). TX: UCC27524 half -> 47 R -> BAV99 crossed diodes -> element (0-15 V square).
33k bleed resets the element after TX. RX: 4.7k + BAV99 clamp (+-0.7 V) + 100 pF (340 kHz RF filter).
OPA1678 stage 1: gain 6, AC-coupled at 800 Hz. Stage 2: MFB band-pass f0 39.8 kHz, Q 2, gain -4. Then 100 R to the ADC.
Element -> ADC +27 dB, pass band 31-51 kHz (sim/reports/rx_chain.md). Band-pass parts 1 % C0G for phase match.

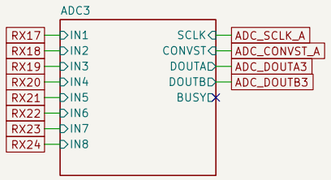


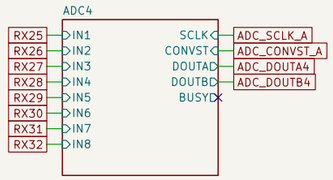
Hybrid 8x8 ultrasonic AESA
Sheet: /Transceivers/CH01/
File: channel.kicad_sch
Title: Transceiver channel (x64): TX, T/R switch, RX amplifier
Size: A4 | Date: 2026-10-20 | Rev: 2
KiCad E.D.A. 10.0.6 | Id: 33/115

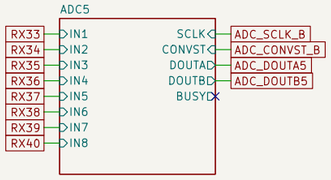
ADCs. ADC k digitises RX 8(k−1)+1 … 8k and returns 2 serial lines (DOUTA/B). ADC1–4 use ADC_SCLK_A/CONVST_A, ADC5–8 the _B outputs.

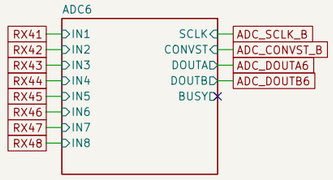


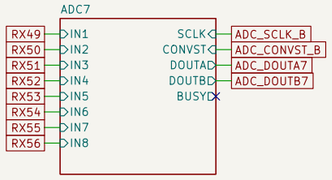


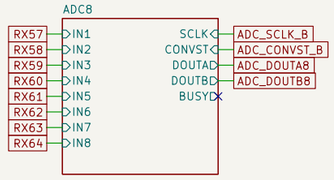












Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/
File: adcs.kicad_sch

Title: ADCs: 8 x AD7606C−16

Size: A3	Date: 2026−10−20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 97/115

ADC: AD7606C-16 (pin-compatible with AD7606B; symbol keeps AD7606 pin names)

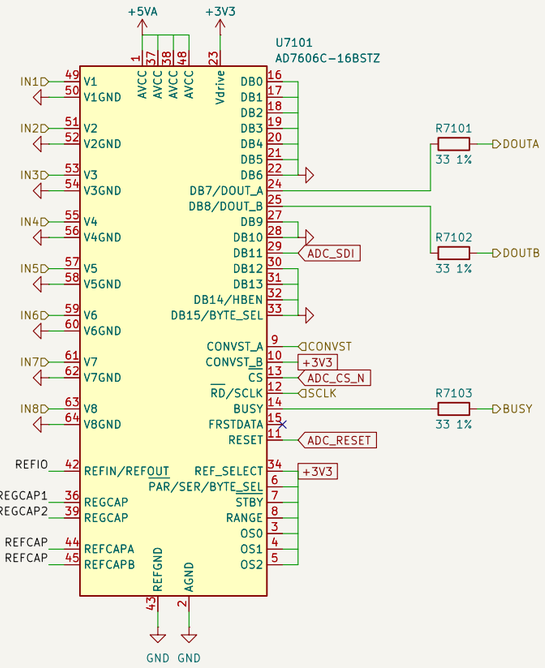
Software mode: OS0-OS2 high. FPGA writes registers over SDI (pin 29, DB11/SDI) at boot:

RANGE_CHx = 0x1 (+-5 V), BANDWIDTH = 0xFF (high bandwidth on all 8 channels), CONFIG DOUT_FORMAT = 2 lines.

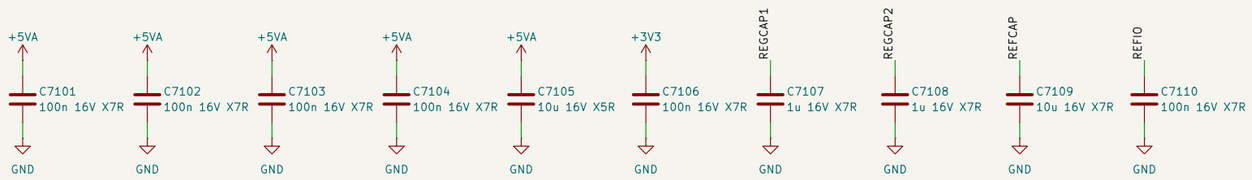
Pin 10 is WR on the B/C parts (parallel register writes only): held high. RANGE pin is ignored in software mode.

33 R source resistors on DOUTA/DOUTB/BUSY (ribbon to the FPGA); BUSY is used from ADC1 only.

DB9/DB10 are tied to GND: never set DOUT_FORMAT to 4 lines. After power-up the FPGA pulses ADC_RESET (full reset), then writes the registers.



Decoupling (place at the pins): four 100 nF on the AVCC pins + 10 uF bulk, 100 nF on VDRIVE; REGCAP 1 uF x2, REFCAP 10 uF, REFIN/OUT 100 nF per datasheet



Hybrid 8x8 ultrasonic AESA		
Sheet: /ADCs/ADC1/		
File: adc.kicad_sch		
Title: ADC (x8): AD7606C-16, software mode, serial 2 DOUT		
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 98/115

MEMS mics. Mics 2k−1 and 2k share MIC_DATAk−1 (SEL high on the odd one). MIC_CLK_A feeds mics 1−4, MIC_CLK_B mics 5−8.

















Hybrid 8x8 ultrasonic AESA

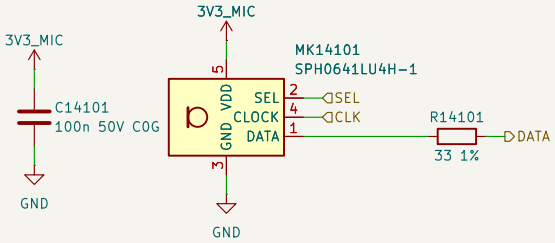
Sheet: /MEMS mics/
File: mics.kicad_sch

Title: MEMS microphones: 8 x SPH0641LU4H−1

Size: A3	Date: 2026−10−20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 106/115

MEMS mic SPH0641LU4H-1 (PDM). Two mics share DATA: SEL high on one, low on the other.

Mounted on the back, sound port through a board hole. 3V3_MIC is the ferrite-filtered rail (power sheet).



Hybrid 8x8 ultrasonic AESA

Sheet: /MEMS mics/MIC1/

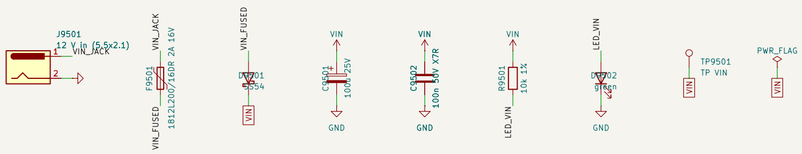
File: mic.kicad_sch

Title: MEMS mic (x8): SPH0641LU4H-1

Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 107/115

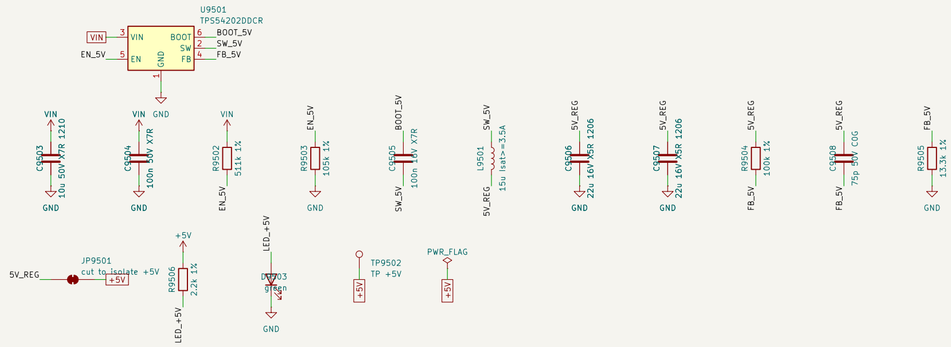
A. 12 V INPUT: jack, polyfuse (16 V), SS54 reverse-polarity diode -> VIN

Use a regulated 12 V adapter only: LM2731 VIN max 14 V. Jack 5.5 x 2.1 mm, >= 1.5 A.



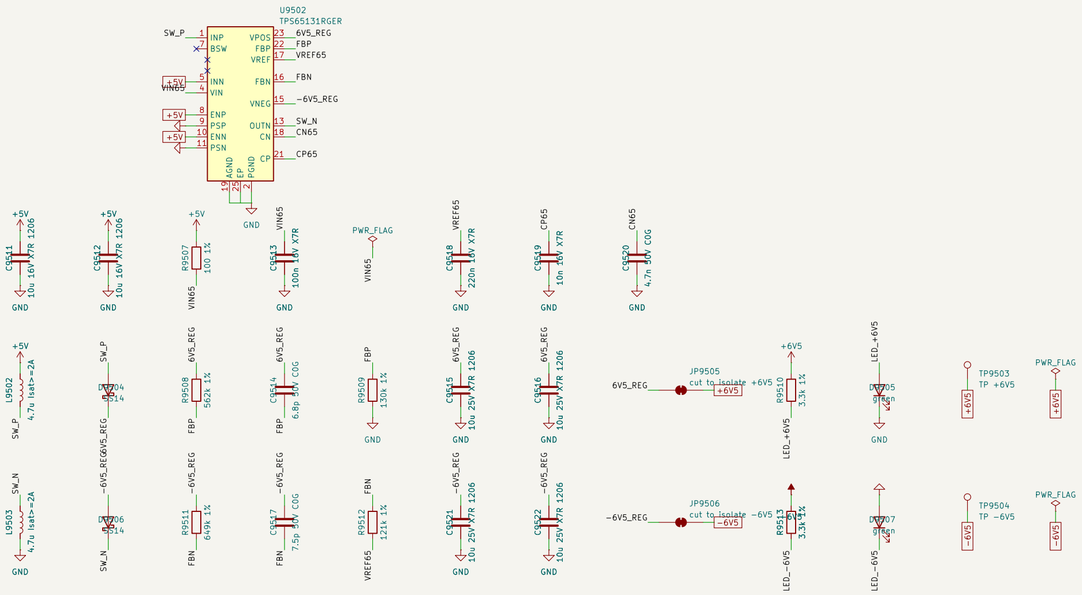
B. +5V: TPS54202 buck (TI reference design), 2 A

Vout = 0.596 V x (1 + 100k/13.3k) = 5.08 V. EN UVLO about 7 V. Load about 1.4 A.



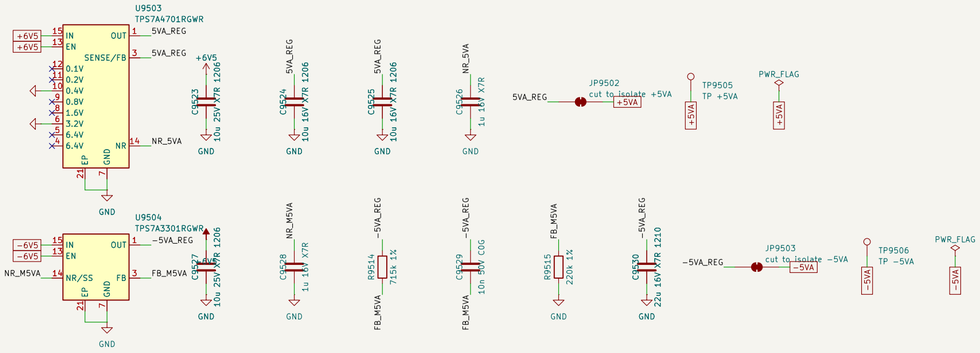
C. +-6V5: TPS65131RGER boost + inverter from +5V (feeds the analog LDOs only)

Drawn with the pin-identical TPS65130 symbol: order TPS65131RGER (2 A switches). PSP/PSN low: fixed-frequency PWM.
+6V5 = 1.213 V x (1 + 562k/130k) = 6.46 V. -6V5 = -1.213 V x 649k/121k = -6.51 V.



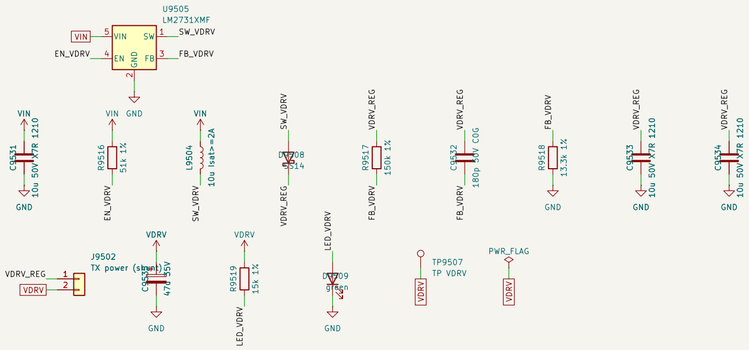
D. ANALOG RAILS: TPS7A4701 +5VA, TPS7A3301 -5VA (low noise, no LEDs)

+5VA: ANY-OUT 1.4 + 3.2 + 0.4 = 5.0 V. -5VA = -1.176 V x (1 + 715k/220k) = -5.00 V.
Load: 64 x OPA1678 (4 mA per channel) + 8 x AD7606C AVCC: about 0.45 A on +5VA, 0.3 A on -5VA.



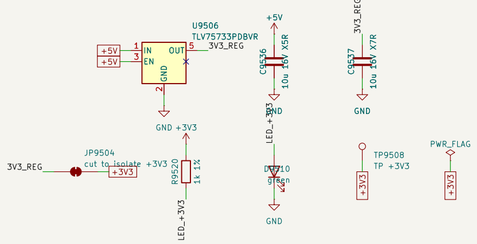
E. VDRV 15 V: LM2731X boost from VIN. Remove the J2 shunt to unpower all TX drivers

1.23 V x (1 + 150k/13.3k) = 15.1 V. Load: 64 elements CW at 40 kHz, CV^2f = 1.3 W, about 0.12 A.



F. +3V3: TLV75733P LDO from +5V (1 A, stable with ceramic caps)

Load < 0.15 A: ADC VDRIVE, clock buffers, logic, mics.

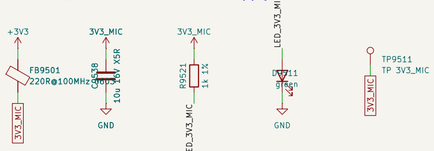


G. GROUND: one GND net and plane

Test points for scope grounds.



H. 3V3_MIC: ferrite-filtered supply for the 8 MEMS mics (remove FB1 to isolate)



Hybrid 8x8 ultrasonic AESA
Sheet: /Power/
File: power.kicad_sch
Title: Power: 12 V in, +5V, +-6V5, +-5VA, VDRV, +3V3, 3V3_MIC
Size: A2 | Date: 2026-10-20 | Rev: 2
KiCad E.D.A. 10.0.6 | Id: 115/115